

The Empirical Claim: A K-Sweep Test (Code KSW)

1 The claim, in one sentence

When a bank recapitalises, how far above the level that triggered the raise it chooses to land is governed by the fixed cost of raising capital, not by the bank’s risk preference — and that relationship has a specific, testable shape.

2 Why this claim, and not another

The theory paper’s actual headline result is that a bank’s asymmetric attitude to capital shortfalls is legible in the thresholds it commits to, not in the volatility of its capital. That headline was the first candidate for an empirical test, and it was set aside — not because it is unresolved, but because when checked, the answer disfavours a direct test of it. Two results from the proofs document establish this:

SCG proves that the *definitional* deficit-to-surplus variance ratio — computed directly from the model, not from data — is a constant of the regulatory cap’s geometry. It carries no information about the preference at all.

- A separate check applies the estimator a real data panel would use to simulated paths from the solved model. The *estimated* ratio does move with the preference, monotonically, but only by about 0.18–0.22 across the full range of the preference parameter, against a per-path standard deviation of 0.16–0.20. The signal is roughly the size of its own noise: usable only aggregated across many banks, not diagnostic from any single one.

That makes the headline claim a poor first empirical target: it needs many banks and a lot of data before it says anything. The K-sweep claim (**KSW**) does not have this problem. Both sides of it — the gap a bank leaves above its trigger, and the fixed cost of raising capital — are in principle observable for a single institution.

3 The claim in detail: two falsifiable predictions

The model has three controls acting on a bank’s capital buffer: a continuously-adjusted risk exposure, dividends paid out at an upper barrier, and lump-sum recapitalisation at a lower trigger. Raising capital costs $K + (1 + \kappa)\xi$ for an injection of size ξ — a fixed component K and a proportional component $\kappa\xi$. Write x_L for the recapitalisation trigger and y_{post} for the capital level the bank raises to. The **gap** is $y_{\text{post}} - x_L$.

Prediction 1 — the gap vanishes at zero fixed cost. In the solved model, at $K = 0$ the trigger and the target coincide exactly (the largest numerical deviation found was 2×10^{-14} , i.e. zero to machine precision). Empirically: a bank facing a negligible fixed cost of raising capital — a private placement to an existing large shareholder, a regulator-arranged injection with no underwriting — should show a post-recapitalisation capital level at or very near the level that triggered the raise, not systematically above it.

Prediction 2 — the gap scales as the cube root of the fixed cost. This is the sharper and more literature-grounded prediction. Classical impulse-control theory with a fixed transaction cost predicts

$$y_{\text{post}} - x_L \propto K^{1/3},$$

a result derived independently of this model, not invented for it. The solved model reproduces it: a log-log regression of the gap against K gives a slope of 0.3845 against the theoretical $1/3$, with the ratio of gap to $K^{1/3}$ stable between 1.26 and 1.35 across a fourfold range of cost values. Empirically: rank recapitalisation episodes by a proxy for the fixed cost involved, and the gap should grow markedly more slowly than the cost — proportional to its cube root, not to the cost itself.

What would falsify this. A robust, non-vanishing gap at near-zero fixed cost; a gap that scales roughly linearly with a fixed-cost proxy rather than sub-linearly; or no detectable clustering of post-recapitalisation capital levels at all, which would be a more basic failure — it would mean the "target level" the model predicts does not describe real recapitalisation behaviour, prior to any question about how that target responds to cost.

4 The data problem, stated honestly

Testing Prediction 2 needs variation in the fixed cost across recapitalisation episodes, and the fixed cost itself is not directly observed. It would be a mistake to assume banks face a large, easily quantified fixed cost: the corporate-finance literature on equity issuance costs for ordinary firms finds the opposite of what the model assumes. Altinkilic and Hansen (2000) find flotation spreads are up to 85% variable cost with *rising* marginal cost as issue size grows — the reverse of a fixed-plus-linear structure. Bühner and Kaserer (2002) find fixed costs account for at most 14–24% of total flotation costs in the German market. Neither result is about bank regulatory-capital raises specifically, which may differ (preferred stock, contingent convertibles, regulatory pre-clearance, or crisis-period government injections could carry a genuinely different cost structure) — but that is a hypothesis, not yet evidence.

The practical implication: proxy for *variation* in cost structure rather than for a fixed cost's dollar value directly. Candidates are issuer size (smaller issuers plausibly bear proportionally larger fixed costs), public versus private/mutual charter status, and whether the episode falls in a stress period (2008–09, 2020) versus ordinary times, since external capital costs are well documented to rise in stress periods regardless of the underlying fixed-cost technology.

5 Data source and what is needed to run this

The source is US bank Call Reports via the FDIC BankFind API — free, public, quarterly, covering every FDIC-insured institution. Two schedules carry what is needed: **RC-R** for regulatory capital

ratios (the buffer), and **RI-A**, “Changes in Bank Equity Capital,” which carries both dividends declared and sale of stock/capital contributions in the same schedule — the exact pair of events (payout, recapitalisation) the model is about.

Concretely, in order:

1. Confirm the relevant RC-R and RI-A fields are reachable and named as expected, and that recapitalisation events (which should be rare, by construction of an impulse control) occur often enough in the data to say anything at all.
2. Pre-register, before looking at results, what counts as a recapitalisation event (a materiality threshold on the stock-issuance line) and what counts as a dividend event, plus which fixed-cost proxy from §4 will be used.
3. Construct the panel: for each recapitalisation event, the capital ratio the quarter before (the trigger level) and the quarter after (the target level); compute the gap.
4. Test Prediction 1: does the gap shrink toward zero as the fixed-cost proxy shrinks?
5. Test Prediction 2: regress $\log(\text{gap})$ on $\log(\text{cost proxy})$; compare the slope to $1/3$, not to 1.

References

- Oya Altinkilic and Robert S. Hansen. Are there economies of scale in underwriting fees? evidence of rising external financing costs. *Review of Financial Studies*, 13(1):191–218, 2000.
- Thomas Bühner and Christoph Kaserer. The structure of external financing costs and the economies of scale view: New evidence from seasoned equity offerings in germany. *European Financial Management*, 8(3):315–358, 2002. doi: 10.1111/1468-036X.00191.